

3.2 Ecosystems

Geography — Eduqas A-Level (A110QS) — Component 3: Contemporary Themes

1. The Ecosystem Concept

An **ecosystem** is a functional unit consisting of a community of living organisms (the **biotic** component) interacting with one another and with their non-living physical environment (the **abiotic** component). The term was coined by the British botanist **Sir Arthur Tansley in 1935** as an explicit replacement for the older, more vague concept of "biotic community", emphasising that life and environment must be analysed together as a single system with measurable inputs, outputs, stores and flows.

Ecosystem: A community of living organisms (plants, animals, microbes) interacting with each other and with their abiotic environment (air, water, soil, climate) as a functional unit. Ecosystems can be analysed at any scale — from a single rotting log to the entire biosphere.

Ecosystems are scale-independent. A small pond, a Welsh hedgerow, the Cairngorms plateau, the Amazon Basin and the entire **biosphere** are all ecosystems — they differ only in size and complexity. Critically, ecosystems are **open systems** with respect to energy (which enters as solar radiation and leaves as heat) but Earth as a whole is essentially **closed** for matter (apart from rare meteoric input and gas escape). This distinction matters because it explains why nutrient *cycles* matter so much — matter must be recycled or it runs out — whereas energy is a *flow* that need only be replenished by the sun.

1.1 Biotic and abiotic components

Component	Examples	Role
Abiotic — climate	Temperature, precipitation, light, humidity, wind	Sets fundamental envelope of what can live where

Abiotic — edaphic	Soil texture, pH, depth, parent material, drainage	Controls plant rooting, nutrient supply
Abiotic — topographic	Altitude, slope, aspect, drainage	Modifies climate locally, affects soil retention
Biotic — producers (autotrophs)	Plants, algae, cyanobacteria, chemosynthetic archaea	Convert solar (or chemical) energy into biomass
Biotic — consumers (heterotrophs)	Herbivores, carnivores, omnivores, parasites	Transfer energy and matter through trophic levels
Biotic — decomposers	Fungi, bacteria, detritivores (worms, beetles)	Recycle dead organic matter, releasing nutrients

Niche: The functional role of a species within its ecosystem — including what it eats, what eats it, when and where it is active, and the abiotic conditions it tolerates. Two species cannot occupy *exactly* the same niche indefinitely (Gause's competitive exclusion principle).

Habitat: The physical place where an organism lives. Often confused with niche — "the niche is the organism's profession; the habitat is its address" (Eugene Odum).

1.2 Ecological interactions

Species do not live in isolation. The principal interactions are:

- **Competition** — for shared limited resources (light, water, nutrients, mates). Intra-specific (within species) is usually fiercer than inter-specific because resource needs are most identical.
- **Predation and herbivory** — consumption of one organism by another. Drives evolutionary arms races (cheetah speed vs gazelle agility; toxin production in plants vs detoxifying enzymes in herbivores).
- **Mutualism** — both partners benefit (e.g. mycorrhizal fungi associating with ~80% of plant root systems, exchanging soil nutrients for plant sugars; pollinators and flowering plants).
- **Commensalism** — one benefits, the other is unaffected (e.g. epiphytes on trees).
- **Parasitism** — one benefits at the host's expense (e.g. mistletoe, tapeworms).

Exam tip — precision matters: Eduqas examiners reward candidates who use the proper terms ("autotroph", "heterotroph", "trophic level", "niche") rather than generic words like "plant" or "place". Aim to introduce a key term in *italics* or bold within the first sentence of any extended answer; it signals A-level command of vocabulary.

2. Energy Flow and Trophic Structure

Energy enters almost every ecosystem as **solar radiation** (the rare exception being deep-sea hydrothermal vent communities driven by chemosynthesis). Approximately 1% of incident sunlight is captured by photosynthesis and stored as chemical energy in plant biomass. From there, energy flows up the food chain through trophic levels.

2.1 Trophic levels

1. **Trophic level 1 — Producers (autotrophs):** photosynthetic plants, algae, cyanobacteria.
2. **Trophic level 2 — Primary consumers (herbivores):** rabbit, caterpillar, zooplankton.
3. **Trophic level 3 — Secondary consumers (primary carnivores):** fox eating rabbit, small fish eating zooplankton.
4. **Trophic level 4 — Tertiary consumers (top carnivores):** golden eagle, killer whale.
5. **Decomposers** — act at every level, returning nutrients to the abiotic store.

2.2 Productivity and the 10% rule

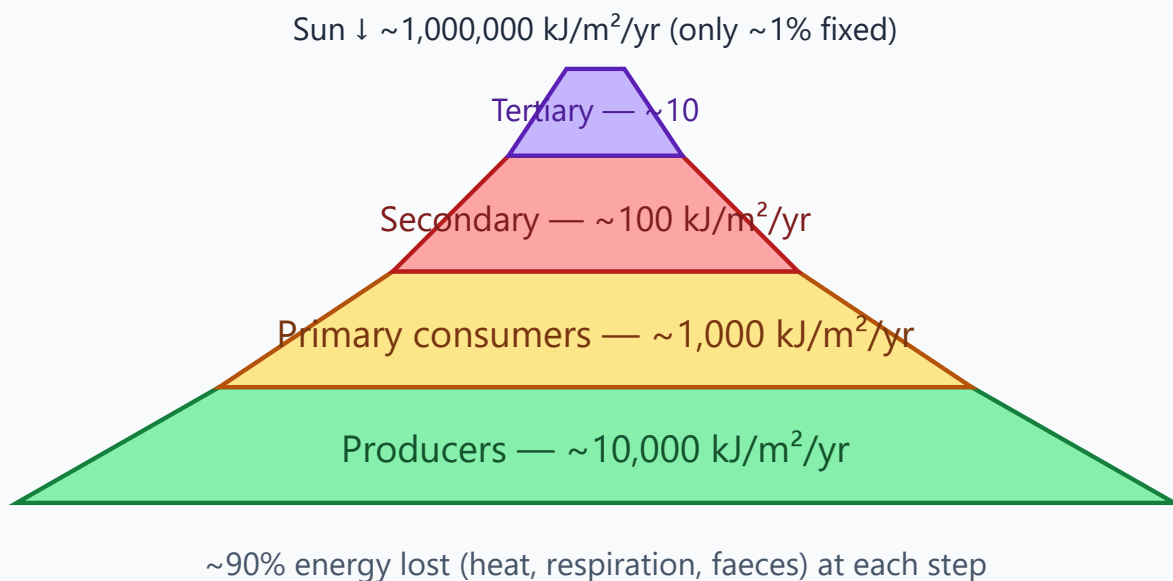
Gross Primary Productivity (GPP): The total rate at which producers fix solar energy into chemical energy through photosynthesis (g C per m² per year, or kJ per m² per year).

Net Primary Productivity (NPP): GPP minus the energy producers use for their own respiration. $NPP = GPP - R$. NPP is the energy actually *available* to higher trophic levels and so is the most useful comparative measure of ecosystem productivity.

An influential rule of thumb is that approximately **10% of the energy entering one trophic level is transferred to the next**. The other ~90% is lost as heat through respiration, in unassimilated material

(faeces, undigested parts), and in tissues never eaten. This **ecological efficiency** explains several major ecological patterns:

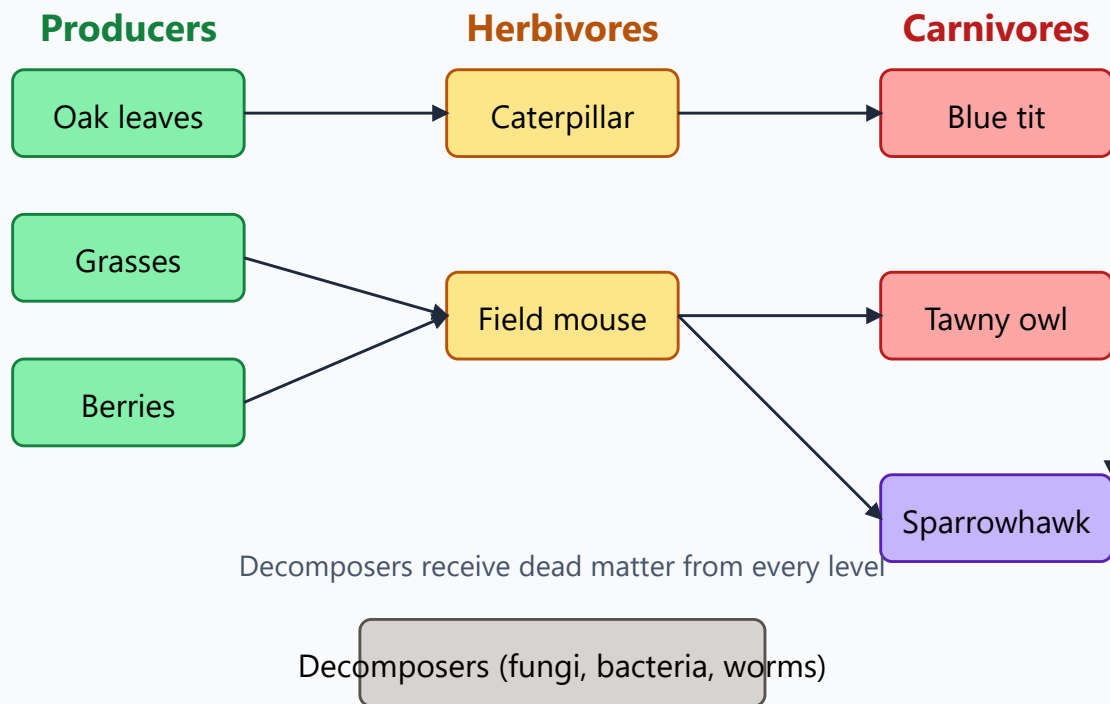
- Top predators are **rare** — very little energy reaches the apex.
- Food chains are typically **short** (4–5 levels maximum) — beyond this insufficient energy remains.
- It is more energy-efficient to feed humans on plants directly than to feed plants to livestock and then eat the livestock — a major argument in food-system sustainability debates.



Trophic pyramid of energy — the ~10% transfer efficiency dictates ecosystem structure.

2.3 Food chains and food webs

A **food chain** traces a single linear path of energy transfer (oak leaf → caterpillar → blue tit → sparrowhawk). A **food web** is the more realistic interlocking network of multiple chains. Food-web complexity is widely regarded as a determinant of **ecosystem stability**: if one species is lost, redundant pathways can compensate. Simplified ecosystems (e.g. monoculture wheat fields) are correspondingly fragile.



Simplified UK woodland food web. Arrows point in the direction of energy flow.

Exam tip — arrows in food webs: Arrows always point from the organism being eaten to the consumer (the direction of energy transfer), not from predator to prey. This is the most common diagram error in 4–6 mark answers.

3. Nutrient Cycles

Whereas energy flows in one direction and is eventually lost as heat, **nutrients cycle** indefinitely between abiotic and biotic stores. The four cycles you must understand are carbon (Topic 2.2), water (Topic 2.1), **nitrogen** and **phosphorus**. Each is presented schematically using the **Gersmehl model** — circles representing stores (biomass, soil, litter), arrows representing fluxes (uptake, decomposition, leaching, runoff). The relative size of each store is highly biome-specific.

3.1 The nitrogen cycle

Nitrogen is a constituent of every amino acid, protein and nucleic acid — biologically essential, but the atmosphere's vast pool of N_2 gas is unusable to most organisms because of the strong triple bond.

Specialised processes convert it:

- **Nitrogen fixation** — converts N_2 to ammonia (NH_3) or ammonium (NH_4^+). Carried out by free-living soil bacteria, by symbiotic *Rhizobium* in the root nodules of legumes (clovers, peas, beans), by cyanobacteria in waterlogged soils, by lightning (atmospheric fixation), and industrially by the **Haber-Bosch process** (200 atm, 450°C, iron catalyst) which provides ~50% of the nitrogen in modern human bodies.
- **Nitrification** — soil bacteria oxidise ammonium to nitrite (NO_2^-) and then to nitrate (NO_3^-), which plants readily absorb.
- **Assimilation** — plants take up nitrate, build amino acids; herbivores ingest plant protein.
- **Ammonification** — decomposers break dead organic matter back to ammonium.
- **Denitrification** — in waterlogged anaerobic soils, bacteria reduce nitrate back to N_2 gas, returning it to the atmosphere.

Human disruption: Industrial fixation via Haber-Bosch produces ~120 million tonnes of fertiliser nitrogen annually, doubling the natural rate of fixation. Excess runoff causes **eutrophication** in rivers and coastal waters — nutrient enrichment leading to algal blooms, oxygen depletion when blooms decompose, and "dead zones" (e.g. the Gulf of Mexico dead zone receives Mississippi-basin agricultural runoff and reaches ~15,000 km² each summer).

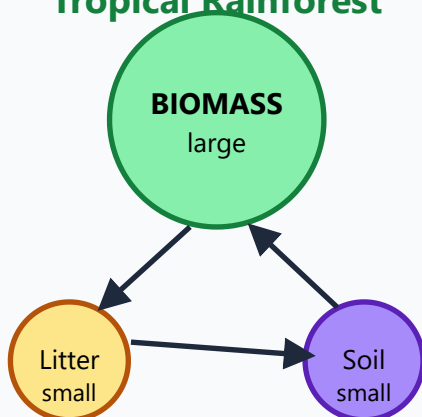
3.2 The phosphorus cycle

Phosphorus, a key constituent of DNA, RNA and ATP, has **no atmospheric component**. The cycle operates over geological timescales: phosphate-bearing rocks weather slowly to release phosphate into soil, plants take it up, animals consume plants, and decomposers return it. Eventually phosphate washes to the oceans and accumulates in sediments, only re-emerging on land when tectonic uplift exposes new rock millions of years later. Most agricultural phosphate fertiliser comes from mining a small number of large deposits (Morocco/Western Sahara holds ~70% of world reserves), raising "peak phosphorus" concerns.

3.3 Gersmehl nutrient cycle diagrams — tropical rainforest vs taiga

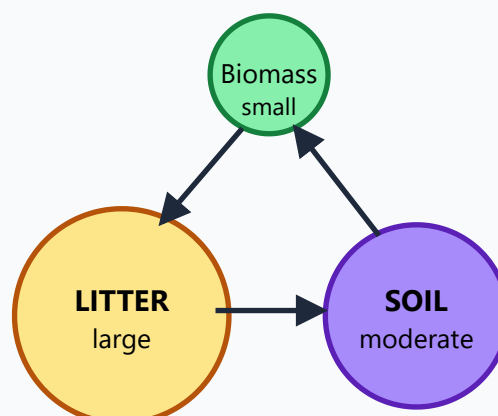
The most important application of Gersmehl diagrams is the contrast between two biomes with radically different storage patterns.

Tropical Rainforest



Rapid recycling, nutrients held in biomass

Taiga (Boreal)



Slow decomposition, nutrients held in litter

Gersmehl nutrient cycle diagrams: rainforest stores nutrients chiefly in biomass; taiga stores them in litter. Circle area is proportional to nutrient store size.

Feature	Tropical rainforest	Boreal forest (taiga)
Climate	Hot, wet year-round (~2000+ mm rain, ~25°C)	Cold winters, short summers (~500 mm, -10 to +15°C)
Decomposition rate	Very rapid (warm + wet)	Very slow (cold + acidic)
Largest nutrient store	Biomass (vegetation)	Litter (slow-decaying needles)
Soil type	Latosol/oxisol — thin, leached, acidic	Podzol — acidic, iron pan
Vulnerability if cleared	Catastrophic — nutrients leach within years	Slow recovery but soil retains some store

Exam tip — rainforest paradox: Tropical rainforests look luxuriant but the *soil* is famously poor. The trees themselves contain almost all the nutrients. Slash-and-burn agriculture works for 2–3 years before yields collapse because the nutrient store has been removed (with the burned biomass) and rainfall leaches what little remains. Always make this paradox explicit in deforestation answers.

4. Succession and Climax Communities

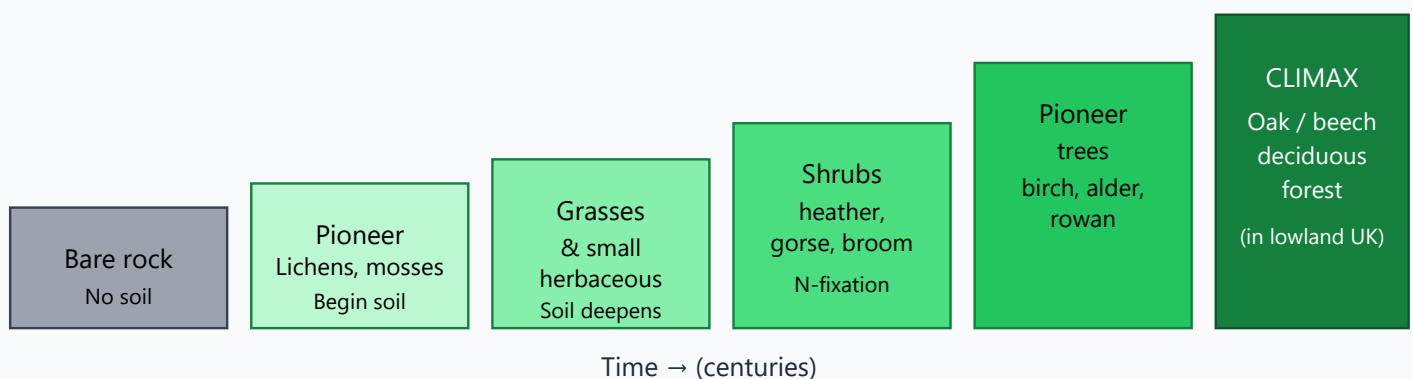
Ecological succession is the directional, non-seasonal change in community composition over time at a particular site, ultimately tending towards a relatively stable end-point known as the **climax community**. The classical theory is associated with Frederic Clements (1916), who portrayed succession as quasi-deterministic. Modern ecology has tempered this with appreciation of stochasticity, alternative stable states and "patch dynamics".

4.1 Primary succession

Primary succession begins on a newly exposed surface that has never previously supported life and lacks soil. Examples include:

- Cooled lava flows (Hawaii, Iceland's Surtsey island after 1963 eruption)
- Land newly exposed by glacier retreat (Glacier Bay, Alaska — classic study by William Cooper from 1916)
- Bare rock left after landslides
- Sand dunes (psammosere) and salt marshes (halosere) — coastal successions

The successional pathway typically passes through these **seral stages**:



Idealised primary succession (lithosere) on bare rock in lowland UK. Each seral stage modifies the abiotic environment, paving the way for the next.

Each seral stage modifies the environment in ways that make it more hospitable to the next: pioneers crumble rock and add organic matter; later stages add depth, retain moisture and shade out the pioneers. This is termed **facilitation**.

Pioneer species: The first species to colonise a previously bare environment. Typically small, fast-growing, tolerant of harsh abiotic conditions, with effective dispersal mechanisms (wind-blown spores, seeds). Lichens are the classic terrestrial pioneers because they can survive on bare rock and accelerate weathering.

Climax community: The relatively stable, self-sustaining end-state of succession in equilibrium with prevailing climate and soil. In lowland UK the natural climatic climax is mixed deciduous oak woodland; in the Scottish Highlands it is Caledonian pine forest; in tropical regions it is rainforest. In practice, almost no part of the UK still supports its true climax because of millennia of human modification.

4.2 Secondary succession

Secondary succession occurs on disturbed land where soil and seed bank remain — e.g. after a wildfire, a hurricane, abandonment of agricultural land, or felling of a forest. Because soil already exists, secondary succession is much faster than primary — recovery to climax forest can take decades rather than centuries.

4.3 Plagioclimax (arrested succession)

Where human activity prevents succession reaching its climatic climax, the resulting community is called a **plagioclimax** (literally "deflected climax"). UK examples are abundant:

- **Chalk grassland** on the South Downs — maintained by sheep grazing and rabbit cropping; remove the grazing and scrub invades, then ash and beech, eventually forming woodland.
- **Heather moorland** in upland Britain — maintained by rotational burning ("muirburn") for red-grouse shoots and sheep grazing. Remove both and heather is replaced by birch and pine.
- **Hay meadows** — biodiverse only because of cutting and grazing regimes.
- **Coppiced woodlands** — sweet chestnut and hazel cycles maintained for centuries.

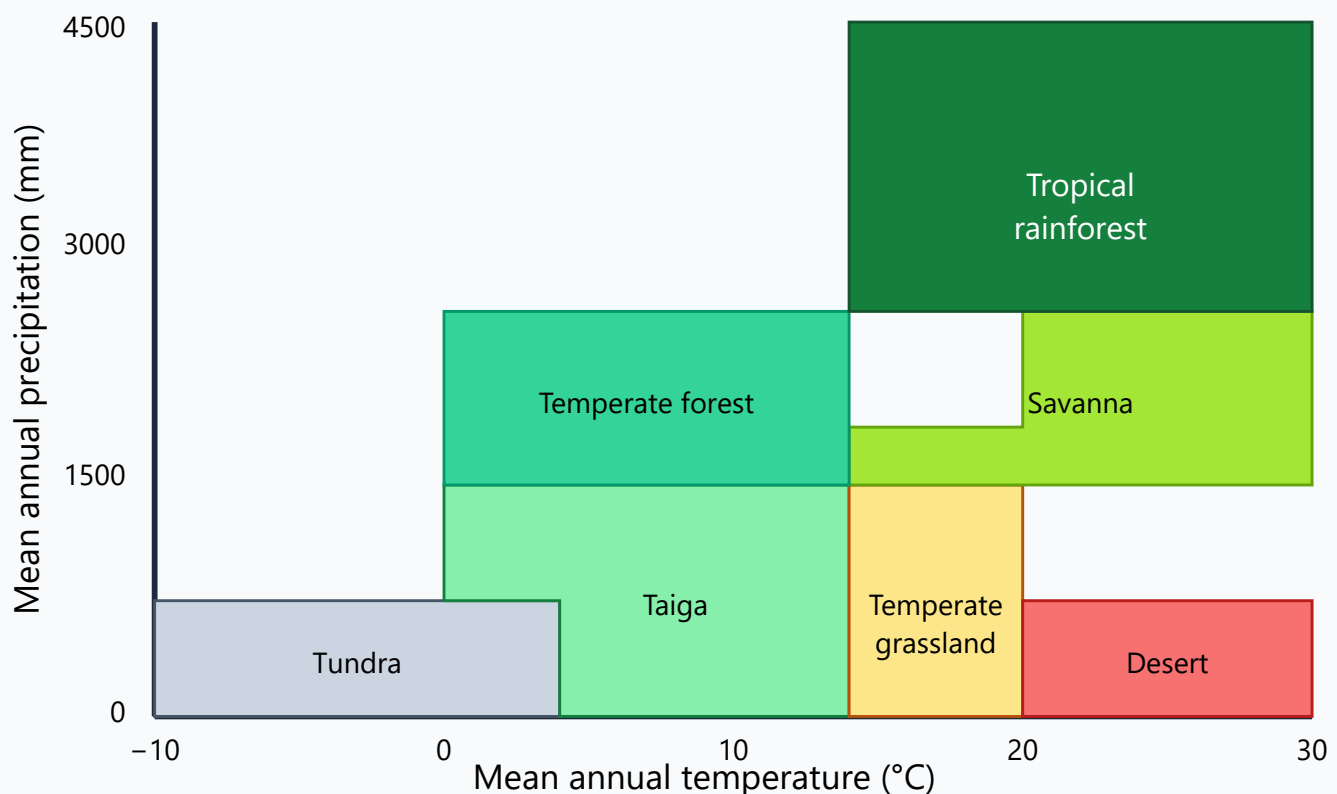
Exam tip — plagioclimax in evaluative answers: When asked whether human management is "good" or "bad" for biodiversity, plagioclimax provides the classic counter-intuitive answer. Many of the UK's most species-rich habitats (chalk grassland, traditional hay meadow) *depend* on continued human disturbance. Stopping the disturbance reduces, not increases, biodiversity.

5. World Biomes

A **biome** is a large-scale community of plants and animals occupying a major habitat over a wide geographic area, characterised principally by the dominant vegetation type that the prevailing climate supports. The Whittaker classification arranges biomes on axes of mean annual temperature and precipitation.

Biome: A continental-scale ecosystem characterised by its dominant vegetation, which is in turn determined chiefly by climate (temperature and precipitation). Each biome has characteristic fauna, soil and ecological processes, regardless of where on Earth it occurs.

5.1 The Whittaker climate-vegetation diagram



Schematic Whittaker diagram: biome distribution on temperature–precipitation axes.

5.2 Major biome characteristics

Biome	Climate	NPP (g/m ² /yr)	Key species & features
Tropical rainforest	~25°C, >2000 mm/yr	2,200	Layered structure (emergent, canopy, understory, forest floor); ~50% of terrestrial species; rapid nutrient cycling in biomass
Savanna	20–30°C, 500–1500 mm seasonal	900	Grasses, scattered trees (acacia, baobab); maintained by fire and grazing (megafauna); wet/dry season
Hot desert	20–30°C, <250 mm/yr	90	Adaptations: succulence (cacti), CAM photosynthesis, deep roots, ephemeral annuals; sparse cover
Mediterranean	Hot dry summers, cool wet winters	700	Sclerophyllous shrubs (chaparral, maquis, garrigue, fynbos); fire-adapted
Temperate grassland	–5 to +25°C, 300–1000 mm	600	Steppes (Russia), prairies (N America), pampas (Argentina); deep fertile soils (chernozem); now mostly arable
Temperate deciduous forest	–5 to +25°C, 750–1500 mm	1,200	Oak, beech, maple; leaf-shedding for winter; brown earth soils; UK's natural climax
Boreal/taiga	–30 to +15°C, 200–750 mm	800	Conifers (pine, spruce, fir, larch); acidic podzol; permafrost in north; covers ~17% of Earth's land
Tundra	–30 to +10°C, <500 mm	140	No trees; mosses, lichens, dwarf shrubs; permafrost; very short growing season

Exam tip — quoting NPP: Examiners reward candidates who use specific NPP values from this table to support comparative claims. "Tropical rainforest NPP (~2,200 g/m²/yr) is roughly fifteen times that of tundra (~140 g/m²/yr) because of year-round warmth, moisture and growing season" is far stronger than "rainforests grow more than tundra".

6. Biodiversity

Biodiversity is shorthand for biological diversity at three nested levels: *genetic* diversity within a species (e.g. crop landrace varieties), *species* diversity within an ecosystem, and *ecosystem* diversity within a region. The Convention on Biological Diversity (1992) is the principal international framework.

Biodiversity: The variety of life at all scales — genetic, species, and ecosystem. Biodiversity underpins ecosystem function, resilience, and the services on which human societies depend.

6.1 Why does biodiversity matter?

- **Ecosystem stability and resilience** — more species means more functional redundancy. If one pollinator declines, others may compensate.
- **Ecosystem services** — biodiversity underpins pollination, decomposition, water purification, climate regulation, all of which have measurable economic value.
- **Option value** — species we have not yet studied may yield future medicines, foods, materials. The Pacific yew gave us the cancer drug paclitaxel; rosy periwinkle gave us vincristine; ~50% of pharmaceuticals derive from natural compounds.
- **Cultural and intrinsic value** — recreation, aesthetic appreciation, religious significance, the simple ethical view that other species have a right to exist.

Measuring biodiversity. Simple measures are *species richness* (the number of species in an area) and *species evenness* (how equally individuals are spread among those species). *Simpson's diversity index* combines both: $D = 1 - \sum n(n-1) / N(N-1)$, where n is the number of individuals of each species and N the total number of individuals; values close to 1 indicate high diversity. Levels of endemism, genetic diversity and the presence of indicator species are also used.

6.2 Biodiversity hotspots

Norman Myers (1988, refined 2000) defined a **biodiversity hotspot** as an area meeting two criteria: at least 1,500 vascular plant species found nowhere else (high **endemism**) *and* already lost $\geq 70\%$ of its primary vegetation (severe threat). There are 36 recognised hotspots covering $< 3\%$ of Earth's land but containing $\sim 50\%$ of plant and $\sim 43\%$ of vertebrate endemics. Examples include Madagascar, the Atlantic Forest of Brazil, the California Floristic Province, the Western Ghats of India, the Sundaland of Indonesia, and the Mediterranean Basin.

6.3 Keystone, indicator and engineer species

Species role	Definition	Examples
Keystone species	Has a disproportionately large effect on its ecosystem relative to its abundance — remove it and the system unravels	Sea otter (controls urchins, allows kelp forests); grey wolf (Yellowstone); elephant; flying fox
Indicator species	Sensitive to environmental change — presence/absence indicates ecosystem health	Lichens (air quality); mayflies (water quality); amphibians (general environment)
Ecosystem engineers	Modify the physical environment, creating habitat for others	Beaver (dams create wetlands); coral (builds reef); earthworm (soil); termite (mounds)
Flagship species	Charismatic, used to attract conservation funding for whole ecosystems	Giant panda; tiger; orangutan; mountain gorilla
Umbrella species	Conservation of which protects many co-occurring species	Jaguar (huge home range); lynx; oak tree (~280 associated insect species in UK)

7. Human Impacts on Ecosystems

The IPBES *Global Assessment* (2019) concluded that human activity has now altered approximately 75% of Earth's land surface, that around 1 million plant and animal species are threatened with extinction, and that current extinction rates are at least tens to hundreds of times higher than the average over the past 10 million years — sufficient to qualify as a "**sixth mass extinction**". The five direct drivers, in order of measured impact, are:

1. **Land/sea-use change** (deforestation, agriculture, urbanisation)
2. **Direct exploitation** (overfishing, hunting, wildlife trade)
3. **Climate change**
4. **Pollution**

5. Invasive alien species

7.1 Deforestation

Tropical forests are being cleared at ~10 million hectares per year (FAO 2020), driven principally by:

- **Cattle ranching** — the single largest driver in the Amazon (~80% of cleared land)
- **Soy** — mostly for animal feed exported to Europe and China
- **Palm oil** — the dominant Indonesian and Malaysian driver
- **Logging** — commercial timber, often illegal
- **Mining** — gold (Amazon, Congo), bauxite, iron, coltan
- **Roads and infrastructure** — act as conduits for further clearance

7.2 Habitat fragmentation

Even when forest is not entirely lost, breaking it into small patches has serious consequences. **Edge effects** — drier, brighter, more wind-exposed forest margins — alter species composition. Reduced patch size cannot support area-demanding species (top predators, large herbivores). Reduced connectivity prevents gene flow, leading to inbreeding. The "50/500 rule" suggests that populations below 50 individuals risk imminent inbreeding depression and below 500 risk long-term loss of evolutionary potential.

7.3 Invasive alien species

Species	Native range	Invasion	Impact
Cane toad (<i>Rhinella marina</i>)	Central/South America	Introduced to Australia 1935 to control sugar-cane beetles	Toxic to native predators (quolls, monitors); has now spread across northern Australia
Grey squirrel (<i>Sciurus carolinensis</i>)	Eastern North America	Introduced to UK ~1876	Outcompetes and carries squirrelepox lethal to native red squirrels
Lionfish (<i>Pterois volitans</i>)	Indo-Pacific	Introduced to Atlantic ~1985 (aquarium release)	No natural predators in Caribbean; devastates reef-fish populations

Japanese knotweed (<i>Reynoutria japonica</i>)	Japan	Brought to UK as ornamental 1850s	Out-competes natives; damages building foundations; eradication ~£1.6bn cost
Burmese python (<i>Python bivittatus</i>)	Southeast Asia	Released into Florida Everglades	Has caused 90%+ decline in some mammal populations in Everglades National Park

Exam tip — invasive species policy: Always distinguish between (a) introduction prevention (biosecurity at borders, ballast-water rules), (b) early detection and rapid response, (c) long-term containment or eradication. Examiners reward candidates who recognise that prevention is far cheaper than cure — New Zealand's biosecurity model is the international gold standard.

7.4 Climate change impacts on ecosystems

- **Range shifts** — species moving polewards/upwards. UK butterflies have shifted ~10 km north per decade.
- **Phenological mismatch** — spring events advancing at different rates. Caterpillars now peak earlier than the chicks that need them.
- **Coral bleaching** — loss of symbiotic zooxanthellae when sea temperature exceeds summer maximum by 1–2°C for several weeks.
- **Polar species loss** — polar bears, walrus, ringed seal dependent on declining sea ice.
- **Biome shifts** — boreal forest expanding into tundra; tundra losing ground; tropical species expanding into temperate zones (e.g. anchovies in North Sea).

7.5 Overexploitation

Direct harvesting beyond the rate of natural replacement: ~33% of marine fish stocks are overfished (FAO 2022); the bushmeat trade is decimating Central African mammals; the global wildlife trade (legal and illegal) is worth ~£15 billion annually and is a major driver of pangolin, rhinoceros, tiger and parrot decline.

8. Conservation Approaches

8.1 Protected areas

The **IUCN protected-area categories** (I–VI) provide an internationally recognised classification:

- **Ia/Ib** — strict nature reserve / wilderness area (no human use)
- **II** — national park (ecosystem protection + recreation)
- **III** — natural monument (specific feature)
- **IV** — habitat/species management area (active management)
- **V** — protected landscape/seascape (people-and-nature; UK national parks fall here)
- **VI** — protected area with sustainable use of natural resources

The **Kunming-Montreal Global Biodiversity Framework (2022)** sets the headline "**30 by 30**" target — 30% of land and 30% of ocean to be effectively protected by 2030. As of 2024, ~17% of land and ~8% of ocean are formally protected, of which a much smaller fraction is "effectively" managed.

8.2 In-situ vs ex-situ conservation

Approach	Definition	Examples	Strengths/Limits
In-situ	Conservation in the species' natural habitat	National parks, MPAs, biosphere reserves, Ramsar sites, ICCAs	Maintains evolutionary processes; requires functional ecosystem; only option for many species
Ex-situ	Conservation outside the natural habitat	Zoos, botanic gardens, captive breeding, gene banks, the Svalbard Global Seed Vault, Kew Millennium Seed Bank	Insurance population; supports reintroduction; cannot recreate ecological processes; expensive per species

8.3 Species reintroduction

- **Grey wolf** reintroduced to Yellowstone 1995 — classic trophic-cascade example (see case study).
- **Eurasian beaver** reintroduced to Britain (Knapdale Scotland 2009; River Otter Devon 2015) after 400-year absence; restoring wetlands, raising biodiversity.
- **California condor** — population fell to 27 birds in 1987; captive breeding then reintroduction; now ~500 wild and captive birds, but lead poisoning from spent ammunition remains a key threat.
- **White-tailed sea eagle** — reintroduced to western Scotland from 1975 (after extinction in the UK in 1918); now ~150 breeding pairs.

- **Lynx (proposed)** — Kielder Forest, Northumberland; ongoing consultation, contested by farming interests.

8.4 Rewilding

Rewilding: A conservation approach that aims to restore self-sustaining natural processes — including the reintroduction of missing species, especially keystone large herbivores and predators — rather than to manage land for a fixed target community. Often allows long-term natural change rather than maintaining a snapshot of past biodiversity.

Rewilding shifts conservation philosophy from *preserving* a particular community to *restoring* the processes that allow nature to determine its own outcomes. Influential UK examples include Knepp Estate (see case study) and the Cairngorms Connect partnership in Scotland (200-year vision for ~60,000 ha). In Europe, Oostvaardersplassen in the Netherlands attempted Pleistocene rewilding using cattle and konik ponies as proxies for aurochs and tarpan; the project drew controversy after winter starvation events but illustrates the principles.

9. Required Case Studies

Case Study 1 — Amazon rainforest: The Amazon basin spans ~5.5 million km² across nine South American countries. It hosts approximately 10% of all known terrestrial species, stores roughly 150–200 GtC of biomass carbon, and generates around half of its own rainfall via evapotranspiration recycling.

Threats: Annual deforestation reached 13,235 km² in Brazilian Amazon in 2021 under President Bolsonaro (a 73% rise from 2018), driven by cattle ranching, soy, illegal gold mining and infrastructure (BR-319 highway). Following the 2022 election of Lula da Silva, monitoring (PRODES system) recorded falls of **22% in 2023 and 31% in 2024** — roughly halving annual clearance from its 2021 peak — demonstrating policy responsiveness.

Tipping point concern: Modelling (Lovejoy & Nobre 2018) suggests cumulative deforestation of ~20–25% would trigger a regime shift to seasonally dry savanna because evapotranspiration recycling would drop below the threshold needed to sustain rainforest climate. Cumulative loss currently sits around 17–18% — uncomfortably close.

Governance: REDD+ payments (e.g. Norway/Germany Amazon Fund — suspended under Bolsonaro, restored 2023); demarcation of Indigenous territories (Yanomami, Kayapo, Xingu) which empirically have the lowest deforestation rates; the 2023 EU Deforestation Regulation (EUDR, applying from December 2026) will prohibit imports of products linked to post-2020 deforestation.

Case Study 2 — Great Barrier Reef bleaching, Australia: The world's largest coral reef system stretches ~2,300 km along Queensland's coast, comprising ~3,000 individual reefs and ~900 islands. Hosts ~1,500 fish species, ~400 coral species. Generates ~AUD \$6.4 billion economic activity annually (tourism, fisheries) and supports ~64,000 jobs.

The bleaching mechanism: Healthy coral lives in symbiosis with photosynthetic algae (zooxanthellae). When sea-surface temperature exceeds the local summer maximum by ~1–2°C for several weeks, the algae are expelled; the coral turns white and may die within weeks if conditions persist.

Mass-bleaching record: Before 1998 mass bleaching events were unknown on the GBR. Since then mass bleaching has occurred in **1998, 2002, 2016, 2017, 2020, 2022 and 2024** — the 2024 event being one of the most severe ever recorded. The 2016–2017 events killed an estimated 50% of shallow-water corals. Recovery requires ~10–15 years between events; the current frequency outpaces it.

Management: Reef 2050 Long-Term Sustainability Plan (Australian + Queensland governments); Crown-of-thorns starfish control programmes; agricultural runoff reduction; Indigenous co-management with Traditional Owner groups; UNESCO has repeatedly considered listing the reef as "in danger". Ultimately, however, bleaching is an emergent property of global CO₂ — without rapid emissions cuts, local management can only delay decline.

Case Study 3 — Knepp Estate rewilding, West Sussex (UK): A 1,400-hectare former arable and dairy estate in the Sussex Weald owned by Charles Burrell and Isabella Tree, who abandoned intensive farming in 2000 after years of failing to make heavy clay soils profitable.

Approach: Removal of internal fences; introduction of free-roaming Tamworth pigs (proxy for wild

boar), longhorn cattle (proxy for aurochs), Exmoor ponies (proxy for tarpan), red and fallow deer. No predetermined target community; let natural processes shape vegetation. Minimal intervention except light monitoring of livestock numbers.

Outcomes (~25 years on): Knepp now supports the UK's largest population of purple emperor butterflies; 2% of the UK's nightingale population (a species in steep national decline); a thriving population of turtle doves (almost extinct nationally); reintroduced white storks now breeding (first UK breeding for ~600 years); and a much-discussed regeneration of ash, oak, willow, blackthorn scrub.

Economics: Revenue from wildlife tourism, glamping, ecologically reared meat and educational programmes is now profitable; supplemented by Higher-Level Stewardship and (post-Brexit) ELM payments. Tree's book *Wilding* (2018) made the project nationally influential and chimed with the post-Brexit policy turn (Agriculture Act 2020) toward "public money for public goods".

Critique: Some neighbouring farmers complain about ragwort and thistle seed; ecologists debate whether the Knepp model is scalable on more productive land; the ethical question of culling (or starvation) when populations exceed carrying capacity remains contested.

Case Study 4 — Yellowstone wolf reintroduction (1995–present), USA: Grey wolves (*Canis lupus*) were extirpated from Yellowstone National Park by 1926 under federal predator-control policy. By the 1990s the park was experiencing severe ecological imbalance: an estimated ~19,000 elk in the northern herd over-browsed willows, aspens and cottonwoods on stream banks, suppressing songbird and beaver populations.

The reintroduction: 31 wolves from Canada were released in 1995–96, with 10 more from Montana in 1997 (41 in total). Within 20 years the population had recovered to ~100 within the park and several hundred in the Greater Yellowstone Ecosystem.

The trophic cascade: Studies by William Ripple and Robert Beschta documented a chain of ecosystem effects: wolves preyed on elk and altered elk behaviour ("ecology of fear" — elk avoided streamside areas where they were vulnerable to ambush). Released from over-browsing, willow and aspen began to regenerate. Beavers, dependent on riparian willow, increased from 1 colony in 1996 to ~9 by 2009. Beaver dams restored wetland habitat, benefiting amphibians, songbirds, fish and

otters. Wolves also suppressed coyote numbers, allowing recovery of pronghorn antelope and red fox.

Caveats: The trophic cascade has been challenged in subsequent papers — some authors argue climate, drought, and grizzly bear population changes also explain elk decline; willow recovery is patchy and varies between drainages. Outside the park, ranchers face livestock losses; wolves were briefly delisted from federal Endangered Species Act protection, allowing state management; ongoing politico-ecological controversy.

Significance: Despite scientific complications, Yellowstone is the world's most-cited example of large-predator reintroduction reshaping a landscape, and influential in current European discussions of lynx, wolf and bear recovery.

Case Study 5 — Borneo orangutans and palm oil: Two orangutan species inhabit Borneo (*Pongo pygmaeus*, ~104,000 individuals, 2016 estimate) and Sumatra (*P. abelii*, ~14,000), both classified Critically Endangered by IUCN. A third species, the Tapanuli orangutan (*P. tapanuliensis*, ~800 individuals), was described in 2017 and is the most endangered great ape on Earth.

Driver: Indonesia and Malaysia together produce ~85% of global palm oil. Plantation expansion has been the principal driver of forest loss in Indonesian Borneo since the 1990s. Estimated forest area has fallen by approximately 50% since 1973 (Gaveau et al., 2014, *PLOS ONE*). Orangutan populations declined ~50% between 1999 and 2015 (Voigt et al., 2018, *Current Biology*). Slash-and-burn clearance of peatland (especially the catastrophic 2015 fires) released immense CO₂ and choked Southeast Asia in haze.

Responses: The Roundtable on Sustainable Palm Oil (RSPO) was founded in 2004 to certify producers meeting environmental and social criteria. Effectiveness debated — certified plantations may still cause indirect deforestation. Indonesia imposed a three-year moratorium on new oil-palm plantation permits (2018–21) and in 2019 made permanent a moratorium on new licences in primary forest and peatland. The European Union **Deforestation Regulation (EUDR, adopted 2023; application postponed to December 2026 for large companies)** will ban EU import of palm oil (and other commodities) from land deforested after Dec 2020 — significant for the world's largest palm-oil consumer market.

Conservation: Borneo Orangutan Survival Foundation rehabilitates orphaned individuals; Sebangau and Tanjung Puting National Parks; corridors connecting fragmented populations; Indigenous Dayak community forestry partnerships.

Synthesis: Palm-oil certification, national moratoria, and consumer-market regulation illustrate the multi-scalar governance required — production country, region, importer market, consumer behaviour and global certification all play a role. No single intervention is sufficient.

Case Study 6 — The Arctic tundra biome (the Eduqas compulsory case study, excluding alpine tundra)

Climate: north of the tree line in northern Canada, Alaska, ice-free Greenland, northern Scandinavia and Russia. Winters are long, dark and severe (often below -30°C); the growing season is only ~50–60 days, with 24-hour daylight in midsummer. Precipitation is low (often <250 mm a year, mostly snow), making it a cold desert, and winds are strong.

Soils: thin, acidic, waterlogged soils over *permafrost*. Only the *active layer* thaws in summer; meltwater cannot drain through the frozen ground, so bogs and ponds form. Cold and waterlogging slow decomposition, so organic matter accumulates — northern permafrost soils hold ~1,500 GtC.

Plants: low-growing, shallow-rooted mosses, lichens, sedges, cotton grass and dwarf shrubs (dwarf willow, dwarf birch, bearberry). Adaptations include cushion shapes and hairy leaves that trap heat and resist wind, dark leaves that absorb sunlight and rapid flowering in the short summer. NPP (~140 g/m²/yr) and biodiversity are low.

Animals: caribou/reindeer (which migrate), musk ox, Arctic fox, Arctic hare, lemmings and snowy owls, with polar bears on coastal sea ice and huge numbers of migratory birds and insects in summer. Adaptations include thick fur and fat, white winter coats, small ears, migration and hibernation.

Interrelationships: permafrost and the short growing season limit plant growth, so herbivores must range widely; lichens are the caribou's winter food; lemming population cycles drive Arctic fox and snowy owl numbers; slow nutrient cycling keeps soils poor; and vegetation insulates the permafrost, so removing it deepens the active layer.

Impacts of climate change: the Arctic is warming about four times faster than the global average. Permafrost thaw causes thermokarst and releases CO₂ and methane (a positive feedback); shrubs spread and the tree line creeps north (“Arctic greening”); sea-ice loss affects polar bears and exposes coasts to erosion; wildfires are more frequent; and rain-on-snow events can lock lichen under ice, starving reindeer (tens of thousands died on the Yamal Peninsula in 2013–14).

Threats: *mineral and fossil-fuel exploitation* — oil at Prudhoe Bay and the Willow project (Alaska), gas on the Yamal Peninsula (Russia), nickel at Norilsk and iron ore at Mary River (Nunavut); pipelines and roads fragment caribou migration routes, and spills are slow to break down (e.g. the 21,000-tonne Norilsk fuel spill, 2020). *Tourism* (Arctic cruises, Svalbard, northern-lights trips) brings disturbance, trampling of slow-growing vegetation and pollution.

Conflicts with indigenous peoples: Inuit, Iñupiat, Gwich’in, Sámi and Nenets depend on hunting, fishing and caribou or reindeer herding. Development brings conflict — e.g. the Gwich’in oppose oil leasing on the coastal plain of Alaska’s Arctic National Wildlife Refuge (the Porcupine caribou herd’s calving ground); in 2021 Norway’s Supreme Court ruled that the Fosen wind farms violated Sámi reindeer herders’ rights; and Nenets herders compete with gas infrastructure on Yamal.

Management strategies: protected areas (Northeast Greenland National Park, the world’s largest national park; the Arctic National Wildlife Refuge); co-management and land-claim agreements that give indigenous peoples a say (e.g. the Nunavut Land Claims Agreement, 1993); the Arctic Council’s environmental working groups, with indigenous organisations as Permanent Participants; engineering that limits damage (the elevated Trans-Alaska Pipeline with caribou crossings); tourism regulation (e.g. on Svalbard); limits on new oil and gas exploration (Greenland stopped issuing new licences in 2021); and, ultimately, global cuts in greenhouse-gas emissions.

10. International Frameworks for Biodiversity

Framework	Year	Purpose
Ramsar Convention	1971	Conservation of wetlands of international importance (~2,500 sites globally)

CITES	1975	Convention on International Trade in Endangered Species — regulates trade in ~38,000 species via Appendix listing
World Heritage Convention	1972	UNESCO designation of natural and cultural sites of outstanding universal value
Convention on Biological Diversity (CBD)	1992 (Rio)	Three goals: conservation, sustainable use, fair benefit-sharing of genetic resources
Aichi Biodiversity Targets	2010–2020	20 targets adopted under CBD; almost all missed by 2020 deadline
Kunming-Montreal Global Biodiversity Framework (GBF)	2022	Successor to Aichi; 23 action targets including the headline "30 by 30" ; pledges of \$200 billion/yr biodiversity finance
IPBES	2012	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services — the biodiversity equivalent of the IPCC

11. Ecosystem Services (MEA 2005)

The **Millennium Ecosystem Assessment (2005)** defined four categories of services that ecosystems provide to people:

Category	Definition	Examples
Provisioning	Material goods extracted from ecosystems	Food (crops, fish, game); fibre (cotton, wood); fuel (firewood, peat); freshwater; medicines (~50% of pharmaceuticals from natural compounds)
Regulating	Benefits from regulation of ecosystem processes	Climate regulation (forests, oceans); flood attenuation (wetlands, floodplains); pollination (insects worth \$235–577 bn/yr); disease regulation; water purification (wetlands)
Cultural	Non-material benefits	Recreation, tourism, aesthetic appreciation, spiritual significance, sense of place, scientific value

Supporting	Underpinning processes that maintain other services	Soil formation; photosynthesis; nutrient cycling; primary production
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Payments for Ecosystem Services (PES) attempts to translate this framework into policy: pay landowners to maintain services rather than convert land for short-term private gain. Examples include REDD+ (forest-carbon payments), the Costa Rica national PES system (paying landowners for water-cycle and biodiversity benefits since 1997), and the EU's Common Agricultural Policy "ecological focus areas".

Exam tip — valuing ecosystem services: When asked to evaluate whether biodiversity loss "matters", structure your answer around the four MEA categories. This converts a vague "we should look after nature" answer into a four-strand argument that examiners will reward, and it allows you to bring in monetary valuations (TEEB, pollinator value, IPBES estimates) where appropriate without sounding like an environmentalist polemic.

12. Future Outlook — The Sixth Mass Extinction?

The five previous mass extinctions in Earth history were caused by asteroid impact, supervolcanism, ocean anoxia and methane release, taking place over thousands to millions of years. The current rate of biological turnover — driven by a single species in less than two centuries — is orders of magnitude faster than the geological background. The IPBES Global Assessment (2019) called for "transformative change": fundamental, system-wide reorganisation across technological, economic and social factors, including paradigms, goals and values, to reduce drivers and put humanity on a sustainable footing.

A particular emerging insight is the **indispensability of Indigenous and Local Communities**: although they constitute <5% of the world's population, they steward areas containing approximately 80% of remaining biodiversity. Many international conservation frameworks (Aichi, GBF) now formally recognise rights-based conservation that places Indigenous land tenure and knowledge alongside scientific approaches.

Exam tip — structuring a 45-mark essay on biodiversity loss: Component 3 option essays are worth 45 marks. A robust answer will:

1. State the scale of loss with figures (1m species at risk, extinction rate tens to hundreds of times the background rate, IPBES 2019).

2. Identify the five direct drivers in order of impact, with one example for each.
3. Use at least two contrasting case studies (e.g. Amazon and GBR) to show different driver-response systems.
4. Evaluate two governance frameworks (e.g. CBD/GBF and EUDR) noting the gap between commitments and outcomes.
5. Reach a synthesised conclusion that recognises both the seriousness of the problem and the policy levers available, including Indigenous stewardship.

13. Common Mistakes to Avoid

- Confusing *habitat* (where) with *niche* (what role).
- Drawing food-web arrows the wrong way (always producer → consumer).
- Treating "biodiversity" as a single number — remember it has genetic, species and ecosystem dimensions.
- Saying tropical rainforest soil is "fertile" — it is famously poor; the trees hold the nutrients.
- Assuming all human management harms biodiversity — many UK plagioclimax habitats depend on continued human disturbance.
- Ignoring scale — succession plays out over decades to centuries; biome change over millennia; mass extinction over centuries-millennia. Match temporal language to the process.
- Quoting Yellowstone wolves uncritically — mention it is contested and has been challenged in subsequent literature.

14. Glossary — Quick Recap

Term	Definition
Autotroph	Organism that produces its own food (usually by photosynthesis)
Heterotroph	Organism that consumes other organisms for energy
NPP	Net Primary Productivity = GPP – respiration; energy available to higher trophic levels

Trophic level	Position in a food chain (producer, primary consumer, etc.)
Pioneer species	First colonisers of bare ground in primary succession
Climax community	Stable end-point of succession in equilibrium with climate/soil
Plagioclimax	Arrested succession maintained by human activity (e.g. chalk grassland)
Biome	Continental-scale ecosystem defined by dominant climate and vegetation
Biodiversity	Variety of life at genetic, species and ecosystem levels
Keystone species	Species with disproportionately large ecosystem effect for its abundance
Hotspot	Area with >1,500 endemic plant species and $\geq 70\%$ habitat loss (Myers 2000)
Endemism	Species found nowhere else in the world
Ecosystem services	Benefits people obtain from ecosystems (MEA 2005: provisioning, regulating, cultural, supporting)
Rewilding	Conservation by restoring natural processes, often via keystone species reintroduction
REDD+	UN mechanism for Reducing Emissions from Deforestation and Forest Degradation
30 by 30	Kunming-Montreal GBF (2022) target: 30% of land and ocean protected by 2030